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Between Shortcut and Ethics: Navigating the Use of Artificial Intelligence in Academic Writing Among Indonesian Doctoral Students

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ABSTRACT

Artificial intelligence (AI) has emerged as a transformative tool in academic writing, leveraging advanced algorithms and natural language processing to significantly improve efficiency, quality and productivity. This study investigates the use of AI tools among Indonesian doctoral students, with a particular focus on ethical standards and their impact on critical thinking. Adopting a phenomenological approach, the research involved 81 participants who provided data through open-ended questionnaires, which were analysed thematically. The findings reveal that AI tools—such as ChatGPT, DeepL, Zotero, Scite AI, Connected Papers and Humata—are extensively used for generating ideas, brainstorming, data analysis, drafting, grammar correction, translation and literature management. While students acknowledge the advantages of AI in enhancing speed and efficiency, they express concerns regarding its ethical implications and potential impact on critical thinking. Key ethical issues include maintaining academic integrity, ensuring manual verification of AI-generated content and avoiding overreliance. The study underscores that, although AI facilitates idea generation and technical tasks, it may undermine critical thinking and analytical skills. To uphold ethical standards and preserve intellectual rigour, the study advocates for a balanced use of AI, positioning it as a complement to, rather than a replacement for, critical scholarly practices.

1 | Introduction

The concerns raised by the Ministry of Research, Technology, and Higher Education (Kemenristekdikti) over the low number of research publications in reputable international journals prompted significant policy interventions. According to the Science and Technology Index (Sinta) from the Ministry of Education, Culture, Research, and Technology, Indonesia's scientific publications from 2008 to 2015 did not exceed 10,000 articles (Amsal 2023). To address this issue, Kemenristekdikti introduced policies aimed at increasing the number of publications

in high-impact journals. To qualify for a professorship, lecturers must meet stringent requirements, including holding a doctoral degree (S-3) and publishing scholarly works as the primary author in reputable international journals. However, the administrative pressure to publish has led to various forms of academic misconduct.

Indonesian lecturers recorded 7639 Scopus-indexed articles published in 2015 and 9454 articles in 2016 (Syahputra 2024). Furthermore, in the period 2019–2021, the number increased to approximately 50,000 articles. It brings Indonesia as the leading

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ASEAN country in terms of scientific output (Amsal 2023). However, this surge in publications has raised concerns about the quality and integrity of the research. Many lecturers may have resorted to shortcuts to meet the publication demands, leading to issues in scholarly publications. Over the past 5 years, the number of retracted or withdrawn journal articles by Indonesian authors has increased significantly, from 13 articles in 2020–2021 to 27 articles in 2023–2024. These retractions were attributed to various reasons, including publisher errors, investigations, self-retractions, duplicate articles, authorship issues, plagiarism, publication in another journal, fake peer reviews, and reference or attribution problems (Adyatama et al. 2024).

The rapid increase in scientific publications in Indonesia can be largely attributed to the integration of technology, particularly artificial intelligence (AI) into academic writing and research practices. The adoption of AI tools has significantly transformed the landscape of scholarly communication, offering both opportunities and challenges that impact academic integrity and the quality of research outputs. AI-powered writing assistants, such as OpenAI's ChatGPT, have revolutionised how researchers conduct literature reviews, draft manuscripts and disseminate knowledge by facilitating the generation of coherent text and streamlining the writing process, thus enhancing productivity (Rahman et al. 2023). It can reduce the time for writing projects and user-friendly nature (Al-Sofi 2024), improved writing skills, and facilitated multilingual and interactive support (Shofiah et al. 2023), so that the convenience provided by AI ignites the intention to adopt AI in their projects (Wang and Reynolds 2024; Wu et al. 2024). However, the works of AI raise critical concerns about the authenticity and originality of scholarly work. Besides, it contained notable mistakes and inaccuracies (Safrai and Orwig 2024) and was not always able to integrate the given information to form meaningful connections between concepts (Kacena et al. 2024).

The ability of AI to produce human-like text can blur the lines between original authorship and machine-generated content, leading to potential ethical dilemmas in academic publishing (Sallam 2023; Schmidt and Meir 2024). For example, the use of AI in generating research papers may inadvertently introduce inaccuracies or superficial content, which could compromise the quality of scholarly discourse (Zielinski et al. 2023). Moreover, reliance on AI tools can foster an over-dependence that may undermine the development of essential research and writing skills among students and early-career researchers.

As noted by Rahman et al. (2023), while AI can assist in various aspects of academic writing, it cannot replace the depth of critical thinking and nuanced analysis required for high-quality scholarly contributions. This shift necessitates a re-evaluation of pedagogical approaches in higher education to ensure that students are equipped with the analytical skills necessary to engage critically with their research (Lin 2023). The challenge lies in harnessing AI's capabilities responsibly, ensuring that it serves as a supplementary tool rather than a crutch that diminishes the quality and integrity of academic work (Schmidt and Meir 2024; Thomas et al. 2023).

Furthermore, the academic community is actively grappling with the implications of AI's integration into scholarly publishing.

The establishment of uniform policies regarding the use of AI in academic publishing is crucial for fostering trust and credibility within the scholarly community (Lund and Naheem 2023). Many leading journals are currently developing guidelines to address the ethical use of AI, emphasising the importance of transparency in disclosing AI's involvement in the research process (DeVilbiss and Roberts 2023; Flanagan et al. 2023). The need for robust ethical guidelines is underscored by the potential for AI-generated content to introduce biases or inaccuracies into published literature, which could have far-reaching consequences for the integrity of scientific communication (Wright et al. 2024; Zielinski et al. 2023).

The ongoing dialogue surrounding AI's impact on academic writing also highlights the necessity for continuous research and policy development to uphold academic standards while leveraging technological advancements effectively. As the landscape of scholarly publishing evolves, institutions must prioritise the cultivation of critical thinking, scholarly rigour and ethical conduct in their educational outcomes (Sullivan et al. 2023). Balancing innovation with the preservation of core academic values is essential to ensure that the benefits of AI integration do not come at the expense of academic integrity (Pividori and Greene 2023; Zielinski et al. 2023).

This research seeks to explore human-AI collaboration in academic contexts, focusing specifically on the experiences and perspectives of doctoral students in Indonesia. It aims to address the following research questions: (1) What AI tools are frequently used by doctoral students, and what are their perspectives on the use of AI in scholarly writing? (2) What are the experiences of doctoral students in Indonesia with AI-assisted writing? (3) How do ethical principles, critical thinking and scholarly rigour influence the use of AI-assisted writing among doctoral students in Indonesia? By examining these questions, this study contributes to the ongoing dialogue about the role of AI in academic writing, highlighting the importance of balancing technological innovation with the preservation of core academic values.

2 | Method

2.1 | Research Design

This study employs a qualitative research method, specifically utilising a phenomenological approach as outlined by Creswell and Creswell (2018). Phenomenology is particularly suited for exploring the lived experiences and perceptions of individuals, making it an appropriate choice for understanding how doctoral students interact with and perceive the use of AI-assisted writing tools in their academic work. By focusing on the shared experiences of the participants, this approach seeks to uncover the essence of their engagement with AI, highlighting the commonalities and differences in their perspectives.

2.2 | Participants

The study involved 81 participants, all of whom share the same characteristic of being doctoral students in educational-related

programmes. These participants were selected to provide a comprehensive understanding of how AI tools are being used in academic writing within this specific demographic. Their insights are valuable in exploring the ethical considerations, challenges and benefits associated with AI in the context of doctoral education.

2.3 | Instrument and Data Collections

Data were collected through open-ended questions distributed via Google Forms. This method was chosen to allow participants the flexibility to express their thoughts and experiences in detail, providing rich qualitative data for analysis. The researcher reached out to participants through WhatsApp groups associated with various scholarship programmes, including the S3 LPDP (Ministry of Finance), BPI (Indonesian Education Scholarships—Ministry of Education and Culture—Ministry of Finance) and BIB (Indonesia Rising Scholarships—Ministry of Religious Affairs—Ministry of Finance). This method of recruitment ensured a diverse yet relevant sample of doctoral students who are engaged in advanced academic work.

2.4 | Data Analysis

The data collected from the open-ended questionnaires were analysed using thematic analysis, a qualitative method that identifies, analyses and reports patterns (themes) within the data. Thematic analysis was selected due to its flexibility and capacity to provide a rich, detailed and nuanced account of participants' experiences. This method aligns with the phenomenological approach adopted in the study, enabling an in-depth exploration of participants' perspectives on the use of AI tools in academic writing. The thematic analysis followed Braun and Clarke (2006) and Braun et al. (2023) six-phase framework, a widely recognised and systematic approach for conducting qualitative data analysis:

1. Familiarisation with the data: The researchers immersed themselves in the data by repeatedly reading the participants' responses to gain a comprehensive understanding of the content.
2. Generating initial codes: Meaningful segments of data relevant to the research questions were systematically coded. The coding process was conducted inductively, allowing themes to emerge from the data rather than being predetermined. To enhance the reliability and validity of the coding process, inter-coder reliability was established. Two researchers independently grouped quotes that expressed the same or similar things in each answer to the research question, then formed a code, followed by the discussion to reconcile any differences. Based on these discussions, the coding scheme was refined to ensure consistency and accuracy. For instance, responses indicating 'using ChatGPT for brainstorming' were coded as 'AI for idea generation'.
3. Searching for themes: The generated codes were organised into potential themes that captured recurring patterns

across the dataset. An iterative and reflexive approach was adopted, with two researchers independently identifying preliminary themes before collaboratively refining them. Related codes, such as 'AI for idea generation', 'AI for grammar correction' and 'AI for literature management', were grouped under broader themes like 'Benefits of AI in Academic Writing'.

4. Reviewing themes: The preliminary themes were critically examined to ensure coherence, distinctiveness and alignment with the data. This involved revisiting the coded data and refining the themes through multiple iterations to accurately represent participants' perspectives.
5. Defining and naming themes: Each theme was clearly defined and labelled to encapsulate its core meaning and relevance to the research objectives.
6. Producing the report: The final phase involved constructing a cohesive narrative that interwove the themes, supported by illustrative participant quotes to provide evidence and enhance the depth of the analysis. To quantify the prevalence of participants' perceptions, the researchers included the number of participant quotes representing the same or similar codes in a table, indicating how many participants contributed to each code. This approach provided a more transparent and comprehensive representation of the data patterns, strengthening the reliability of the findings.

2.5 | Ethical Considerations

The study adhered to rigorous ethical standards to ensure the rights, privacy and well-being of the 81 doctoral student participants. Informed consent was obtained through a detailed consent form embedded in the Google Forms survey. The form explicitly outlined the research purpose, which aimed to explore the use of AI tools in academic writing within doctoral education. It provided comprehensive information about participants' roles, emphasising their voluntary participation, the confidentiality of their responses and their right to withdraw at any time without consequence.

To mitigate ethical concerns related to plagiarism, authorship and the potential loss of critical thinking, the survey questions were carefully designed to prompt participants to reflect on both the advantages and challenges of using AI tools in academic writing. This approach encouraged participants to provide nuanced and balanced perspectives rather than endorsing AI use uncritically. Additionally, participants were informed that the study sought to understand their experiences and perceptions rather than promote or legitimise AI tools for academic writing.

Participant recruitment via WhatsApp groups associated with reputable scholarship programs was conducted in a respectful and non-coercive manner. To protect participants' privacy, all data collected through Google Forms were anonymized by removing any personally identifiable information. The findings were reported in aggregate, maintaining participants' confidentiality throughout the research process. The researchers

transparently communicated the potential implications of the study, including how the results could contribute to the development of ethical guidelines and institutional policies on AI usage in academic settings.

3 | Results

3.1 | Experiences of Using AI in Academic Writing

The integration of AI into academic writing represents a significant paradigm shift in higher education, challenging traditional notions of authorship, creativity and intellectual rigour. As educational practices continue to evolve in the 21st century, the reliance on AI tools underscores a complex interplay between human cognition and machine learning, where the boundaries between human and artificial contributions to scholarly work become increasingly blurred. From a constructivist perspective, this evolution marks a critical juncture in the way knowledge is constructed and disseminated. AI's role in academic writing is not merely a matter of enhancing efficiency or refining outputs; it raises profound questions about the nature of learning itself.

AI tools have become indispensable for doctoral students, particularly in supporting various aspects of writing such as drafting, grammar correction and translation. Table 1 displays the experiences of doctoral students in using AI for academic writing tasks. The ability to expedite the writing process, enabling quicker production of drafts and improved language accuracy, is highly valued. However, this reliance on AI for technical aspects of writing raises significant concerns about the erosion of students' linguistic skills and their ability to maintain these skills independently. The convenience provided by AI may inadvertently lead to a diminished capacity for students to engage deeply with the mechanics of language, which is crucial for academic rigour.

The role of AI in literature search and management is profound, with tools like Google Scholar and Scite AI significantly streamlining the process of finding relevant literature, managing references and verifying citations. This has led to enhanced research efficiency, particularly in tasks like systematic literature reviews (SLRs), which can now be completed in shorter timeframes. However, the data also suggest a potential downside: the dependence on AI for such critical tasks could weaken students' research skills and their ability to critically engage with the literature. This reliance on AI might reduce the depth of their understanding and their capacity to perform thorough, independent literature analysis.

The use of AI for generating and developing research ideas is another prominent theme. Tools like ChatGPT have proven particularly helpful when students struggle with formulating ideas or connecting theoretical concepts. While AI provides a valuable crutch for brainstorming and overcoming writer's block, there is a growing concern that over-reliance on AI for creative thinking could stifle the development of original, critical thought. The fear is that students might become dependent on AI for intellectual stimulation, thereby compromising their ability to generate innovative ideas independently.

One of the most appreciated features of AI tools is their ability to enhance efficiency and productivity. By accelerating various research and writing processes, AI enables students to manage larger volumes of work more effectively. However, this efficiency comes with a potential cost: the risk of students becoming overly dependent on AI, which could erode their intrinsic motivation and their ability to manage research tasks autonomously. The ease provided by AI might lead to a superficial approach to complex academic tasks, undermining the depth and rigour that are essential to scholarly work.

Despite the clear benefits, there are notable concerns about the quality of AI outputs and their alignment with academic standards. Students express apprehension that AI-generated content may not always meet the required academic rigour, raising fears that reliance on AI might compromise the authenticity and integrity of their work. Furthermore, the simplification of complex academic tasks by AI tools poses a risk of promoting a shallow understanding of the material, which could detract from the educational experience and the development of critical analytical skills.

A recurring theme in the data is the need for caution in relying too heavily on AI. While students acknowledge the convenience and efficiency that AI tools provide, they are also acutely aware of the potential drawbacks, such as reduced critical thinking, creativity and the ability to perform independent analysis. The metaphor of 'opening Pandora's box' aptly captures the sentiment that, while AI offers considerable advantages, it also carries hidden risks that could lead to a generation of researchers lacking deep analytical skills and independent thought. The data suggests a growing awareness among students of the need to balance the use of AI with the cultivation of their intellectual capacities, ensuring that AI serves as a tool for enhancement rather than a substitute for critical academic engagement.

We visualised the most frequently mentioned concept related to the use of AI in academic writing with key terms like 'AI', 'writing', 'assistance', 'critical thinking', 'efficiency', 'reliance' and 'creativity' in Figure 1. This visually reinforces the main themes discussed in the analysis, emphasising the balance between the benefits and potential drawbacks of AI use.

'AI' and 'writing' are naturally dominant in the word cloud, reflecting the central focus of the discussion on how artificial intelligence is being integrated into the academic writing process. 'Assistance' highlights the supportive role that AI tools play in the academic writing process. This term suggests that students are primarily using AI as a tool to aid their writing, helping to streamline tasks such as grammar correction, translation and literature management.

'Critical thinking' stands out as a crucial concept in the word cloud, indicating a major concern related to the use of AI. The analysis emphasises the potential risks of over-reliance on AI, particularly the fear that it could erode students' critical thinking skills.

'Efficiency' is another key term that appears prominently. The word cloud visually reinforces the idea that AI significantly

TABLE 1 | Kinds of experience of using AI in academic writing.

Category	Code	Frequency	Sample Quotes
AI for writing assistance	Drafting and Writing	15	‘These tools are very helpful, especially for making the initial draft faster and correcting mistakes I might have missed.’ ‘Hemingway Editor helps make my writing more concise and clearer, although it sometimes suggests removing parts that are actually important academically.’
	Grammar Correction	12	‘I started using AI tools like Grammarly and Hemingway Editor to improve my grammar and enhance my writing style.’
	Translation	18	‘Mostly used for translating articles.’
	Technical Tools	9	‘I needed time to adapt and understand their features. Especially for more technical tools like Python and R in data analysis.’
AI for literature search and management	Literature Search and Management	14	‘I also use Google Scholar to search for relevant literature, which saves me a lot of time compared to manual searches.’ ‘With AI, I can complete two SLR research projects in one month.’
			‘AI tools like Scite AI are effective in verifying citations, but I still need to double-check to ensure accuracy.’
AI for idea generation	Idea Generation	10	‘AI tools like ChatGPT help me when I’m struggling to find initial ideas, although further development is still needed to match my research topic.’ ‘When I’m having trouble finding ideas and difficulty starting the first sentence, I usually ask for references from AI and then develop them further. AI is a helpful partner in writing or research, but that doesn’t mean we fully rely on AI.’
			‘Before AI, I had difficulty putting research ideas into words and connecting them with relevant theories.’
			‘AI can provide effectiveness and time efficiency in searching for references on the research theme being conducted.’ ‘Without AI, it would take much longer for doctoral students to graduate.’
AI for efficiency and productivity	Increased Efficiency	11	‘It can improve the efficiency, quality, and speed of writing by providing various features that support writing, from simple tasks like grammar checking to more complex ones like generating content or structuring documents.’
Concerns with AI	Concerns and Limitations	12	‘AI does simplify the writing process, but sometimes the results do not meet the academic standards I desire.’ ‘That’s why you have to master various educational theories, so you can verify the mistakes generated by AI.’
			‘Ideas and data, at the very least, must be ours. The analysis can be a combination of our own and AI’s.’
			‘Humans must have a unique critical side that results from a complex understanding of facts and theories. AI doesn’t have that.’

(Continues)

TABLE 1 | (Continued)

Category	Code	Frequency	Sample Quotes
Caution in reliance	Caution in Reliance	13	‘AI helps me write articles, but I feel I need to be cautious not to rely too much on this technology.’ ‘Using AI too often makes me dependent. Even for answering questions in class discussions, I use AI.’ ‘Everything is AI: answering questions, writing research proposals, writing papers, scientific articles. This is dangerous.’ ‘It’s like opening Pandora’s box. All difficulties are solved, whether it’s finding ideas, narrating findings, analysing data, or excellent translation. But remember, ease creates a weaker generation.’ ‘Wow, I’m dependent on AI.’

enhances the efficiency of academic writing by speeding up various processes, such as literature search, drafting and editing. However, this term also ties into the theme of ‘reliance’, which appears frequently and points to the potential drawback of becoming too dependent on AI tools. The analysis suggests that while efficiency is a major benefit, there is a need to be cautious about how this efficiency might affect students’ overall learning and skill development.

‘Creativity’ is another significant term, highlighting concerns that AI might impact the creative aspects of academic writing. The visualisation suggests that creativity is a key consideration for students when using AI tools, with the analysis exploring the potential trade-offs between leveraging AI for idea generation and the risk of diminishing students’ capacity for original, critical thought.

Additionally, the concept map in Figure 2 illustrates the relationships between these themes, showing how different aspects of AI use—such as drafting, grammar correction, idea generation and literature management—interact with concerns about reliance, critical thinking and academic integrity.

3.2 | Commonly Used AI Tools in Academic Writing

AI tools designed to enhance various aspects of the writing process range from basic grammar and style checkers to advanced applications that assist with data analysis, citation management and even content generation. The growing reliance on AI in academia underscores the increasing demand for efficiency, accuracy and innovation in scholarly work. This section delves into the most commonly used AI tools in academic writing, analysing their functionalities, benefits and the crucial role they play in upholding the intellectual rigour of academic outputs.

Table 2 shows a list of the top 12 AI tools frequently used by doctoral students in academic writing tasks. ChatGPT

emerges as the most frequently employed AI tool, particularly valued for its versatility in idea generation. Students rely on it not only for brainstorming but also for answering questions, conducting analyses and translating text. This widespread use underscores a significant shift towards AI handling complex, multifaceted tasks that traditionally demanded considerable human effort. The increasing dependence on ChatGPT reflects a broader trend in academia where efficiency and versatility are highly prized.

While ChatGPT is favoured for its broad applicability, DeepL and Grammarly are specifically utilised for translation and grammar correction, respectively. DeepL’s adjustable style makes it a preferred choice for accurate translations, while Grammarly’s reputation for precision in grammar correction remains strong. However, the gradual shift of some users towards ChatGPT for these tasks suggests a growing preference for comprehensive tools that can streamline multiple functions within a single platform, signalling a potential consolidation of AI tools in academic writing.

Zotero and Mendeley remain indispensable for citation management, reflecting their entrenched role in academic writing. The emergence of Scite AI, a tool aiding in the crafting of discussion sections, highlights an increasing demand for AI that transcends basic citation management and contributes more actively to the writing process. This shift points to a future where AI’s role in academic writing becomes more integrative, potentially influencing not just the management but also the content and structure of scholarly work.

Frequent use of tools like Connected Papers and Litmaps, which are designed for mapping literature and visualising research trends, underscores their critical importance in academic research. These tools enable students to identify gaps in the literature, a crucial step in crafting original research contributions. Their prevalence in the dataset highlights the growing recognition of the need for AI to assist in navigating and synthesising vast bodies of research, thus facilitating more informed and innovative academic writing.



FIGURE 1 | Word cloud visualisation related to AI in academic writing.

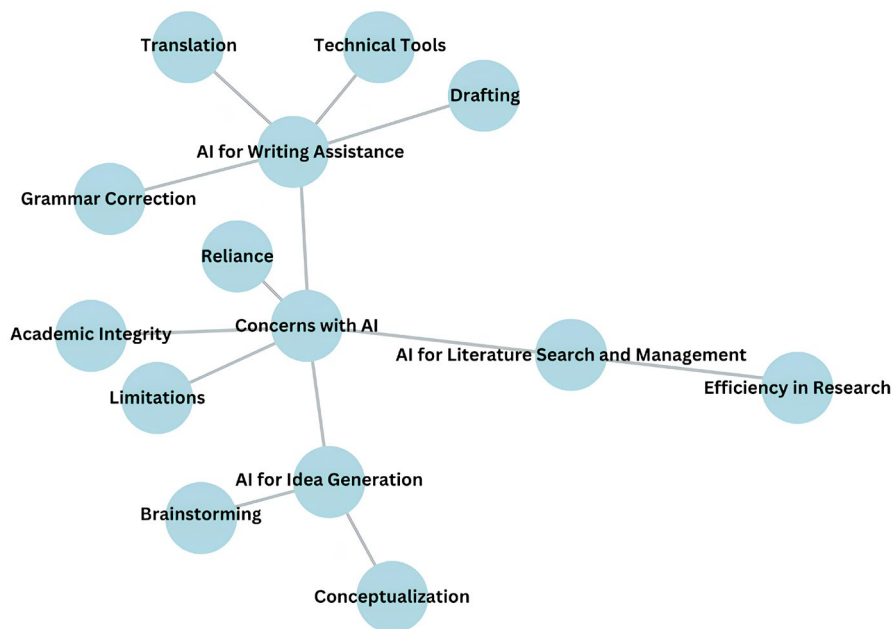


FIGURE 2 | Relationship map related to AI in academic writing.

Tools such as Elicit and SciSpace are particularly valued for their ability to simplify complex academic texts and summarise extensive literature. The reliance on these tools reflects the significant pressure on doctoral students to manage large volumes of information efficiently. By providing substantial time savings, these tools allow students to focus more on critical analysis and synthesis, rather than merely processing information, thereby enhancing the overall quality and depth of their academic work.

Figure 3 Visualises the tools and concepts related to the use of AI tools in academic writing, reflecting their widespread

adoption among students. ChatGPT emerges as the most prominent tool, underscoring its versatility in tasks like idea generation, drafting and translation. Other essential tools such as DeepL, Grammarly, Zotero, Mendeley and Scite AI also feature prominently, indicating their crucial roles in translation, grammar correction and literature management. The core functions of idea generation, writing, translation and grammar correction illustrate the reliance on AI to support fundamental aspects of academic writing, while functions like literature management, summarisation and simplification highlight AI's role in organising research and making complex tasks more efficient. Despite the benefits, terms like caution in reliance and concerns and

TABLE 2 | Commonly used AI in academic writing.

AI Tool	Usage Category	Frequency	Sample Quotes
ChatGPT	Idea Generation	30	‘ChatGPT seems like it can do everything. Answer questions, create analyses, brainstorm ideas, find relevance between theories, categorize data—basically a lot.’
	Translation and Writing		‘ChatGPT is also great for translation. It recognizes the terms used in education.’
DeepL	Translation	28	‘For translation, I use DeepL because the translation style can be adjusted.’
Grammarly	Writing (Grammar Correction)		‘I used to use Grammarly to check article translations, but it turns out ChatGPT is more flexible.’
Zotero	Literature Management	20	‘Zotero and Mendeley are very useful for storing citations and automatically creating bibliographies.’
Mendeley			‘You must have Zotero or Mendeley; manually citing is outdated.’
			‘Zotero and Mendeley are very useful for storing citations and automatically creating bibliographies.’
Scite AI	Literature Management	20	‘I recently discovered a new AI called Scite AI. It not only provides citations for the paragraphs you paste, but it can also help craft the discussion section in an article.’
Connected Papers	Literature Mapping and Analysis	15	‘Connected Papers makes it easier for me to find related literature and map research networks.’
Litmaps	Literature Mapping	15	‘Before conducting research, we need to find theoretical gaps—leave it to Connected Papers or Litmaps.’
	Visualising Research Trends	6	‘Since using Litmaps, it has become much easier for me to develop a strong theoretical framework, visualizing research trends and identify theoretical gaps.’
Elicit	Summarization and Simplification	12	‘Elicit and Scispace help me understand and simplify complex academic texts.’
Scispace			‘PhD students are often tasked with summarizing comprehensive books. One course can require summarizing 5–6 books. Thankfully, Elicit is here, so I don’t have to read all the books and strain my eyes.’
	Summarization and Simplification	12	‘SciSpace AI can generate summaries and highlights from research articles, making it easier for me to grasp core concepts and findings. I no longer need to read numerous articles, which can sometimes be overwhelming, especially when confronted with extensive data and numbers.’
Genei	Brainstorming and Outlining	8	‘Nowadays, SLR are highly popular in published articles, as they do not require field data. Additionally, many of these articles are easier to review, offering concise summaries and highlighting key points with tools like Genei.’
Humata		8	‘Just ask Humata, if you want to know the contents of the article’

TABLE 3 | Categorisation of ethical principles in using AI for academic writing.

Category	Code	Frequency	Sample Quotes
Ethical use of AI	Manual Verification and Adjustment	18	'I always manually verify the results or suggestions provided by AI.'
	Academic Honesty and Integrity	7	'Peer review from experts is very much needed. If a paper is suspected of being made by AI, not just assisted by AI, it can be directly rejected.'
			'Many are starting to see AI as an instant solution to writing problems, but this clearly risks honesty and academic integrity.'
			'Indonesia ranks second in publications in predatory journals, meaning what? Even the literacy to select journals is poor, let alone publishing quality papers. This academic climate needs to be improved.'
	Transparency in AI Use	6	'As long as data transparency exists, using AI as an assistant is not a problem.'
	Limiting AI to Technical Assistance	9	'AI should only be used for technical issues in writing, like translation, citation, and data classification; beyond that, it's prohibited.'
Concerns about AI overuse	Overreliance on AI	7	'Many are tempted to overuse AI, even to the point of writing entire articles, due to high publication pressure.'
	Weakening Mastery of Theory and Critical Analysis	18	'If everything is done by AI, when do we master theories and analyze facts based on the theories we have mastered? If not continuously honed and handed over to AI, critical analysis skills will decline. This awareness is something I always reinforce within myself.'
Ethical challenges in academia	Publication Pressure	12	'The pressure to publish in reputable journals tempts many students, including me, to use AI unethically.'
			'Our problem lies between rules, workload, and ethics. If the first two squeeze, ethics can be defeated.'
	Strengthening Authorship Position	11	'Oh, don't ask about ethics; many act as ghostwriters and seek help from ghostwriters in writing and publication.'
			'This ethics issue shows the complexity of bullying in Indonesia's academic tradition, where those in power or senior easily insert names in papers without any contribution, just for money.'
	Plagiarism and Fabrication Risks	10	'AI is often used to avoid hard work in research, producing writings that are not truly original.'
			'With high publication demands, many end up using AI to falsify research results. I've also fabricated data.'
	Strengthening the Role of Academic Journals	6	'Journals must be the guardians of ethical issues, ensuring the review process is conducted honestly and credibly, so AI-made papers are rejected.'
			'I'm thinking of one solution, where if any paper is eligible to be published in a reputable journal, the author must present it online, so the author's capacity as an expert in their research is proven.'

(Continues)

TABLE 3 | (Continued)

Category	Code	Frequency	Sample Quotes
Preventive measures	Laws and Regulations	16	‘There must be laws that regulate. If caught plagiarizing or fabricating data, it should impact their academic position.’ ‘Ethics return to personal responsibility. It’s hard to regulate. There must be strict laws.’
Supportive academic environment	Excessive Workload	8	‘Doctoral programs are the highest level of education, leading to becoming a professor. The rules require publication in Scopus journals. Many are not capable, and as proof, Indonesia is the second-largest publisher in predatory journals.’ ‘The workload of lecturers in Indonesia is overwhelming. Besides the Tridharma (the three pillars of higher education), there are additional responsibilities such as managing study programs or specific units, accreditation forms, and other administrative tasks. On top of that, there is the obligation to publish every semester. Ultimately, they turn to AI.’
	Need for a Rigorous Scholarly Community	5	‘There must be a community that fosters the right academic environment. This ensures that before becoming a professor, individuals can learn from the publication experiences of truly competent professors.’ ‘Ideally, there should be a community that ensures the articles submitted for professorship go through a presentation process and are tested for their scholarly expertise.’



FIGURE 4 | Word cloud visualisation of ethical principles in using AI.

critical thinking, as overreliance on AI could potentially erode this essential skill. The presence of regulations in the word cloud suggests a call for stricter guidelines to govern AI use, indicating a recognised need for formal frameworks to protect academic integrity and ensure that ethical standards are upheld in the face of these pressures.

3.4 | Impact of AI-Assisted Writing Tools on Doctoral Students' Critical Thinking and Scientific Rigour

The use of AI as a cognitive tool invites a re-examination of what it means to engage deeply with academic content, challenging students to navigate the delicate balance between leveraging AI for

intellectual advancement and maintaining their critical autonomy as scholars. Table 4 shows the impacts of using AI in academic writing. It highlights the potential risks of over-reliance on technology, where the mechanisation of thought processes could inadvertently undermine the development of essential academic skills such as critical thinking, original analysis and nuanced argumentation. The dual-edged potential of AI in education necessitates a critical pedagogical approach that not only integrates AI into the learning process but also rigorously interrogates its influence on the epistemological foundations of academic inquiry. As educators and scholars grapple with the implications of AI in academic writing, there is an urgent need to foster a pedagogical environment that encourages reflective engagement with AI tools.

We visualised the most frequently occurring terms from the dataset in Figure 5. It offers a visual emphasis on the key themes and concerns expressed by the students. Critical Thinking stands out prominently, indicating that students are deeply concerned about preserving their capacity for deep and analytical thought in an academic environment increasingly influenced by AI. This term underscores the anxiety that AI might erode the essential skill of critical thinking, a cornerstone of scholarly work.

AI is central to the discussion, frequently appearing as students grapple with its dual nature in academic writing. While they acknowledge the efficiency and support AI offers, the term over-reliance emerges as a significant concern, reflecting students' fears that excessive dependence on AI could stifle their intellectual development. This worry is closely linked to creativity,

TABLE 4 | Impact of AI in academic writing.

Category	Code	Frequency	Sample Quotes
Impact on critical thinking	Diminished Critical Thinking	38	'I fear that I might start losing the ability to think deeply because I'm relying too much on this tool.'
	Overreliance on AI	16	'The more I use AI, the more I depend on it, and it worries me about my thinking ability.'
	Erosion of Analytical Skills	17	'I'm afraid my analytical skills are becoming dull because AI is doing most of the work.'
Limiting AI use as an assistant tool	AI Aids Creativity	27	'AI helps me find ideas quickly, but honestly, I feel like something is missing—the critical thinking process.'
	Need for Manual Verification	15	'I always manually check the results provided by AI, ensuring data accuracy and strong arguments.'
	AI as an Aid, Not a Replacement	7	'As long as the use is limited to idea generation and framework formation, it seems acceptable.'
	Speed and Efficiency	6	'AI makes technical tasks easier, but the more I use it, the more I depend on it.'

**FIGURE 5** | Word cloud of the most frequent term in AI for academic writing.

which appears as both a benefit and a potential drawback of AI use; students recognise AI's potential to enhance creativity but are wary of its impact on critical thinking.

Verification is another key term, highlighting the importance students place on manually checking AI-generated outputs to ensure they meet academic standards and maintain scientific rigour. Finally, efficiency frequently appears, indicating students' appreciation for how AI can streamline their work processes. However, this is tempered by a cautious awareness of AI's broader implications, particularly its potential to diminish the depth and quality of their intellectual engagement.

4 | Discussion

This phenomenological study tried to unpack the doctoral students' experiences and perspectives in using AI for conducting their scholarly writing project. The study's findings enrich the theoretical understanding of the use of AI in academic settings by highlighting the academic values and students' experiences and perceptions in the writing project. The integration of AI in academic writing has indeed transformed the landscape of

scholarly communication, presenting both remarkable advantages and significant ethical dilemmas. Tools, such as ChatGPT, enhance clarity and assist in idea generation, respectively, yet they also raise critical concerns regarding academic integrity and the potential erosion of essential cognitive skills. For instance, Hemingway Editor's suggestion to remove content for brevity can inadvertently lead to the omission of vital arguments, thereby compromising the depth of academic discourse (Buriak et al. 2023). Similarly, while ChatGPT can facilitate brainstorming and alleviate writer's block, it cannot replicate the nuanced development of research ideas that are fundamental to scholarly work (Dergaa et al. 2023).

This study revealed that ChatGPT stands out for its versatility in idea generation, drafting and translation. At the same time, tools like DeepL, Grammarly, Zotero, Mendeley and Scite AI play key roles in translation, grammar correction and literature management. AI supports academic writing through idea generation, writing and language refinement while also aiding research organisation through summarisation and simplification, illustrating the reliance on AI to support basic skills and simplify complex tasks in academic writing. It indicates that the reliance on AI tools may foster a culture of superficial engagement with content, as users may become overly dependent on these technologies for generating ideas rather than cultivating their analytical capabilities (Dwivedi et al. 2023); whereas, AI still had significant inaccuracies and errors (Kacena et al. 2024; Safrai and Orwig 2024) that raise academic concerns, such as plagiarism, creativity and critical thinking (Al-Sofi 2024). It highlights that AI cannot be used as a stand-alone tool but needs human verification to grab the benefits of AI for improving students' critical thinking and enhancing students' social, cognitive and emotional engagement (Guo and Wang 2025; Liu and Wang 2024).

The use of AI in literature management, exemplified by tools like Scite AI and Zotero, streamlines citation processes and enhances the accuracy of references. However, this automation

risks diminishing scholars' engagement with the intricacies of citation practices, potentially leading to a lack of understanding of the ethical implications surrounding authorship and intellectual property (Hastuti and Syafruddin 2023). While the automation of bibliographic tasks increases efficiency, it may also inadvertently contribute to a decline in critical thinking skills, as scholars might rely on AI to navigate complex citation networks without fully grasping the underlying principles of scholarly communication (Ryan and Stahl 2021). In other words, the tendency to overreliance on AI can result in carelessness about its limitations leading to academic dishonesty and decreased academic writing skills (Al-Sofi 2024).

This study also confirmed that students perceived AI tools are valued for expediting writing, improving language accuracy, enhancing research efficiency and allowing students to handle larger workloads. They assist in literature management, idea generation and boosting productivity. These findings corroborate the findings of Al-Sofi (2024), Kacena et al. (2024) and Shofiah et al. (2023), who confirmed that AI tools are perceived as useful and promising for conducting academic tasks, thus predicting students are more likely to have a higher intention to use AI (Wang and Reynolds 2024; Wu et al. 2024). However, students still had concerns about overreliance and fears of the impact on their critical and analytical thinking, academic rigour and AI output quality.

It indicates that students are still aware of the risk of adopting AI in their work. Besides, Wu et al. (2024) found that perceived usefulness did not affect attitude or intention to use AI. Students would stop using AI if they experience difficulties or dissatisfaction.

The ethical considerations surrounding AI usage in academia are further complicated by the pressures of the 'publish or perish' culture, which can lead to unethical practices such as data fabrication and plagiarism (Nam and Bai 2023). The ethical landscape of AI in academia necessitates a careful balance between leveraging technological advancements and maintaining academic integrity. AI should be positioned as a supplement, not a replacement (Al-Sofi 2024), and evaluating AI responses is essential to prevent misinformation and inaccuracies (Safraï and Orwig 2024). Zhang et al. (2023) highlighted the integration of ethics into AI curricula is crucial for preparing future scholars to navigate the ethical challenges posed by AI technologies. This is particularly pertinent in light of the increasing prevalence of AI-generated content, which raises questions about authorship and the authenticity of academic work (Dergaa et al. 2023). The need for transparency in AI usage is paramount; ethical guidelines should delineate the acceptable boundaries of AI assistance, ensuring that it serves as a complement to human intellect rather than a substitute (Lin 2023).

The prevalence of predatory publishing among Indonesian academics highlights a critical issue within the academic community, where the pressure to publish often overshadows the importance of ethical practices and quality scholarship. Research indicates that a significant portion of Indonesian scholars engage with predatory journals, which compromise the integrity of academic output and undermine the credibility of research (Latan et al. 2023). The systemic issues contributing

to this phenomenon include a lack of information literacy and inadequate guidance in selecting reputable journals, which exacerbate the vulnerability of researchers to exploitative publishing practices (Wiratningsih 2018). Furthermore, the regulatory framework in Indonesia, which prioritises quantity over quality in academic publications, has inadvertently fostered an environment conducive to such unethical practices (Pourret et al. 2021). This situation underscores the urgent need for a robust academic community that emphasises ethical standards and promotes critical engagement with scholarly work.

The misuse of artificial intelligence (AI) in academic writing further complicates the landscape of academic integrity. AI tools, while offering efficiencies in generating content, can also serve as shortcuts that diminish the originality and integrity of scholarly work. The reliance on AI for writing can lead to a culture of complacency, where researchers may resort to automated solutions rather than engaging deeply with their research (Miao et al. 2023). This concern is compounded by the pressures of high publication rates, which can drive unethical behaviours in academic writing such as fabrication and falsification of data and images, and plagiarism (Nazarovets 2024). The integration of AI in academic contexts necessitates stringent regulations and accountability measures to mitigate these risks and uphold the standards of academic integrity (Uddin and Abu 2024).

Moreover, ethical issues surrounding authorship and publication practices are prevalent in academia, with ghostwriting and academic bullying being significant concerns. Senior academics often exploit their positions to claim authorship on papers without contributing meaningfully, undermining the integrity of academic authorship (Himawan 2021). This exploitation calls for a cultural shift within academic institutions to promote transparency and accountability in research practices. While individual responsibility is crucial, institutional support and clear guidelines are essential to foster an environment that prioritises ethical scholarship (Bozkurt 2024).

The dual-edged nature of AI integration in academic work presents both opportunities and challenges. On one hand, AI can enhance efficiency and creativity, aiding in tasks such as literature review and data analysis (Slimi and Villarejo-Carballido 2024). On the other hand, there is a growing anxiety that over-reliance on AI tools may erode critical thinking and analytical skills among researchers (William 2024). The need for manual verification of AI-generated outputs emphasises the importance of maintaining rigorous intellectual engagement with content, as AI cannot replace the depth of analysis required for high-quality scholarship (Tang et al. 2024). This nuanced perspective advocates for a balanced approach to AI use, where it serves as a complement to human intellect rather than a substitute (Rodrigues et al. 2024).

5 | Conclusions

5.1 | Research Findings

Academic writing is an indicator of one's creativity and competence. The use of AI can be an effective tool to help doctoral students produce higher-quality academic writing. Our study

explores the most frequently used AI tools by doctoral students for their academic writing tasks, their experiences and the influence of ethical principles, critical thinking and scholarly rigour in academic writing tasks. It revealed that ChatGPT, DeepL, Zotero, Mendeley, Scite AI, Connected Papers, Litmaps, Elicit, SciSpace, Genei and Humata are the most popular AI tools among doctoral students. They frequently used those AI tools for various academic writing tasks such as generating ideas, brainstorming, outlining, drafting, translating, managing literature, summarising, mapping literature and visualisation. They perceived that AI tools are very helpful, enhance the quality of writing and save time. The study also reveals a complex relationship between AI-assisted writing tools and academic integrity among doctoral students. Students recognise the tools' benefits but express concerns about overreliance and potential ethical breaches.

5.2 | Practical Implications

While AI can be helpful, human oversight is crucial to maintain academic rigour. Therefore, on the practical level, our findings suggest that a set of practical guidelines on AI use in academic settings is essential, especially in academic writing. It can support the academic environment to enhance the quality of students' writing and prevent unethical practices.

5.3 | Research Limitation

Despite these contributions, this study had some limitations. Firstly, while the study focuses on experiences and perspectives for using AI on academic writing projects, it does not deeply explore specific cases in AI-assisted academic writing such as authorship, plagiarism and transparency. Lastly, it also does not focus on students' critical thinking skills and overreliance on using AI.

5.4 | Future Research Directions

Future studies should explore the ethical dilemmas posed by AI in academic writing, including authorship, plagiarism and transparency. Moreover, future research should also assess the impacts of using AI on students' critical thinking skills in literature synthesis and argument development in writing.

Author Contributions

Hardiyanti Pratiwi: conceptualization, methodology, data collection, thematic analysis, writing – original draft preparation, and correspondence. **Suherman:** data analysis, validation, visualization and review of manuscript drafts. **Hasruddin:** literature review, validation, and critical review of the discussion and ethical considerations. **Muhammad Ridha:** supervision, editing – review and revision, and final approval of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

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